1. (2 puntos) Calcula y simplifica:

a) $\dfrac{2}{\frac{1}{6}+\frac{1}{2}} - \dfrac{3}{1+\frac{1}{2}} = \dfrac{2}{\frac{1}{6}+\frac{3}{6}} - \dfrac{3}{\frac{2}{2}+\frac{1}{2}} = \dfrac{2}{\frac{4}{6}} - \dfrac{3}{\frac{3}{2}} = \dfrac{2}{\frac{2}{3}} - \dfrac{3}{\frac{3}{2}} = \dfrac{6}{2} - \dfrac{6}{3} = \dfrac{18}{6} - \dfrac{12}{6} = \dfrac{6}{6} = 1$

b) $\dfrac{\frac{1}{2}:\frac{5}{4}+\frac{3}{5}}{\frac{1}{6}+\frac{4}{3}\cdot\frac{3}{5}} = \dfrac{\frac{2}{5}+\frac{3}{5}}{\frac{1}{6}+\frac{4}{5}} = \dfrac{\frac{5}{5}}{\frac{5}{30}+\frac{24}{30}} = \dfrac{1}{\frac{29}{30}} = \dfrac{30}{29}$

2. (1 punto) Calcula el valor numérico del polinomio P (x) = 3xy + x^2y – (1/2)y + 5, para x = –1 e y = 2

$P(-1, 2) = 3 \cdot (-1) \cdot 2 + (-1)^2 \cdot 2 - (1/2) \cdot 2 + 5 = -6 + 2 - 1 + 5 = -1 + 1 = 0$

3. (3 puntos) Deja en forma de potencia de exponente positivo y calcula:

a) $\left[\left(\frac{3}{4}\right)^5 \cdot \left(\frac{3}{4}\right)^2\right]^{-2} = \left[\left(\frac{3}{4}\right)^7\right]^{-2} = \left(\frac{3}{4}\right)^{-14} = \left(\frac{4}{3}\right)^{14}$

b) $\left(\frac{2}{3}-\frac{3}{4}\right)^{-2} \cdot \left(\frac{1}{3}-\frac{7}{9}\right)^{-1} = \left(\frac{8}{12}-\frac{9}{12}\right)^{-2} \cdot \left(\frac{3}{9}-\frac{7}{9}\right)^{-1} = \left(\frac{-1}{12}\right)^{-2} \cdot \left(\frac{-4}{9}\right)^{-1} = \left(\frac{12}{1}\right)^2 \cdot \left(\frac{-9}{4}\right)^1 =$

$= -(3 \cdot 2^2)^2 \cdot \dfrac{3^2}{2^2} = -\dfrac{3^2 \cdot 2^4 \cdot 3^2}{2^2} = -3^4 \cdot 2^2$

c) $\dfrac{2^4 \cdot 3^4 \cdot 12^2 \cdot 3^{-3}}{10^2 \cdot \left(\frac{1}{3}\right)^{-2} \cdot \left(\frac{1}{5}\right)^2 \cdot (2^2)^3} = \dfrac{2^4 \cdot 3^4 \cdot (3 \cdot 2^2)^2 \cdot 3^{-3}}{(5 \cdot 2)^2 \cdot 3^2 \cdot 5^{-2} \cdot 2^6} = \dfrac{2^4 \cdot 3^4 \cdot 3^2 \cdot 2^4 \cdot 3^{-3}}{5^2 \cdot 2^2 \cdot 3^2 \cdot 5^{-2} \cdot 2^6} = \dfrac{2^8 \cdot 3^3}{2^8 \cdot 3^2} = 3$

4. (1 punto) Calcula:

$$4\sqrt{18} - 2\sqrt{8} - 3\sqrt{32} = 4\sqrt{3^2 \cdot 2} - 2\sqrt{2^3} - 3\sqrt{2^5} = 4 \cdot 3\sqrt{2} - 2 \cdot 2\sqrt{2} - 3 \cdot 2^2\sqrt{2}$$
$$= 12\sqrt{2} - 4\sqrt{2} - 12\sqrt{2} = -4\sqrt{2}$$

5. (1 punto) Efectúa la siguiente operación y simplifica:

$$6 \cdot \left[\frac{3(x-5)}{2} - \frac{y-x}{2} - \frac{y}{6} - 3\right] = 6 \cdot \left[\frac{3 \cdot 3(x-5)}{6} - \frac{3(y-x)}{6} - \frac{y}{6} - \frac{18}{6}\right]$$
$$= 9(x-5) - 3(y-x) - y - 18 =$$
$$= 9x - 45 - 3y + 3x - y - 18 = 12x - 4y - 63$$

6. (2 puntos) Factoriza y simplifica:

$$\frac{x^4-5x^3+6x^2+4x-8}{x^2-x-2}=\frac{(x+1)\cdot(x-2)\cdot(x-2)\cdot(x-2)}{(x+1)\cdot(x-2)}=(x-2)\cdot(x-2)$$

$$=(x-2)^2$$

	1	–5	6	4	–8
–1		–1	6	–12	8
	1	–6	12	–8	0
2		2	–8	8	
	1	–4	4	0	
2		2	–4		
	1	–2	0		

	1	–1	–2
–1		–1	2
	1	–2	0